Tarea 5.2

1. Creamos la primera red denominada “Red1”:

docker network create --driver bridge --subnet 172.28.0.0/16 --gateway 172.28.0.1 red1

```
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel> docker network create --driver brid
4ed283bf7b2a83949b4da948c572dc7b571413577a40ac95eb2ab74e0c17cf23
```

Esto crea la red con los siguientes datos:

- **Nombre: redil**
- **Driver: bridge**
- **Dirección de red: 172.28.0.0**
- **Máscara: 255.255.0.0**
- **Gateway: 172.28.0.1**

2. Procedemos a crear la segunda red la cual denominaremos “Red2”:

docker network create --driver bridge red2

```
Administrador: Windows PowerShell
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel> docker network create --driver bridge red2
470019fce9d2729d30fb3f9ff56dda35347b70b532f892b3750ef2c0d550fa8a
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel>
```

3. Comprobamos que efectivamente se han creado:

docker network ls

```
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel> docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
a404c96076ca   bridge    bridge    local
88b11cd0748a   host      host      local
bdfcb81501fc   none      null      local
4ed283bf7b2a   red1      bridge    local
470019fce9d2   red2      bridge    local
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel>
```

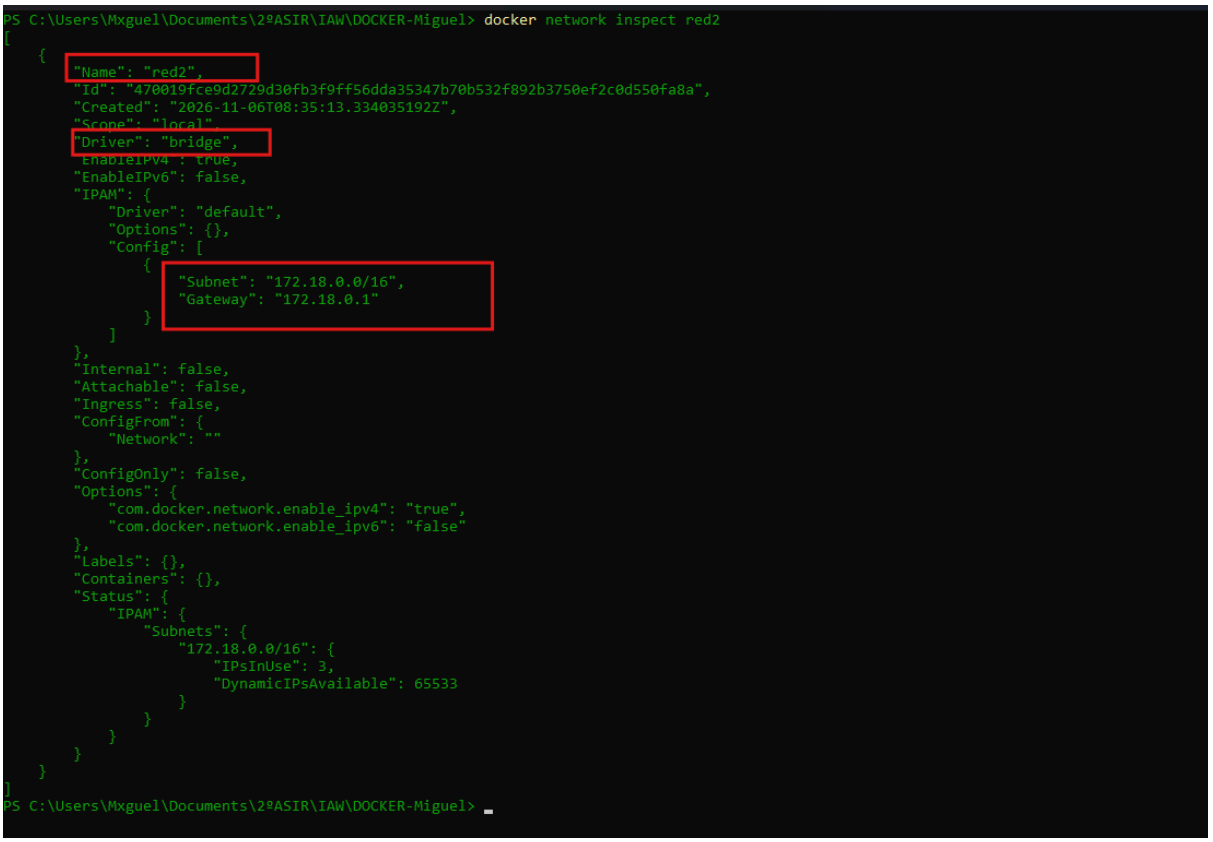
4. Inspeccionamos la red 1 y vemos la configuración que establecimos:

docker network inspect red1

```
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel> docker network inspect red1
[
    {
        "Name": "red1",
        "Id": "4ed283bf7b2a83949b4da948c572dc7b571413577a40ac95eb2ab74e0c17cf23",
        "Created": "2026-11-06T08:33:59.9419887Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {},
        "Containers": {},
        "Status": {
            "IPAM": {
                "Subnets": {
                    "172.28.0.0/16": {
                        "IPsInUse": 3,
                        "DynamicIPsAvailable": 65533
                    }
                }
            }
        }
    }
]
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel>
```

5. Inspeccionamos la red 2 y vemos la configuración asignada por defecto:

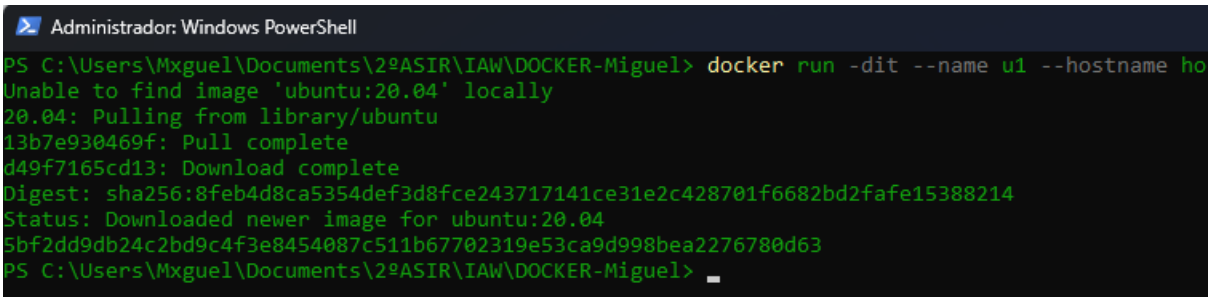
docker network inspect red2



Tarea 5.3

1. Comenzamos creando un contenedor (u1) poniendo en ejecución un contenedor de la imagen ubuntu:20.04:

docker run -dit --name u1 --hostname host1 --network red1 --ip 172.28.0.10 ubuntu:20.04



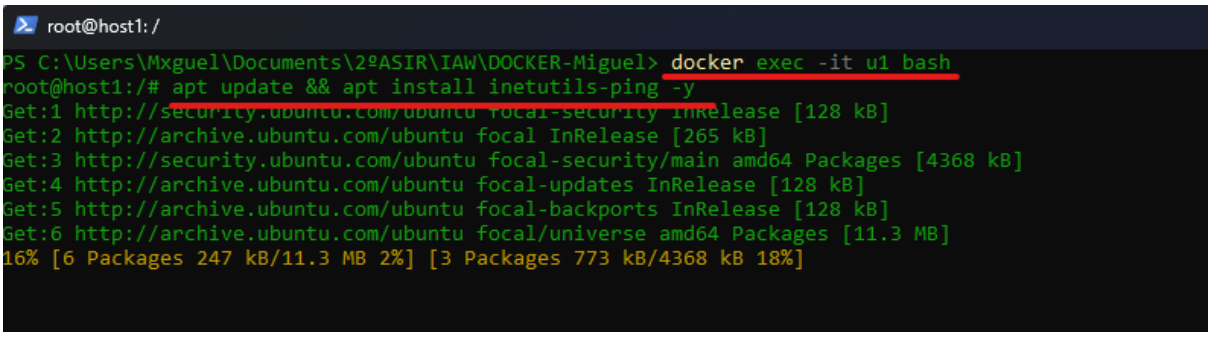
2. Instalamos ping en el contenedor u1:

1º Lo ejecutamos:

docker exec -it u1 bash

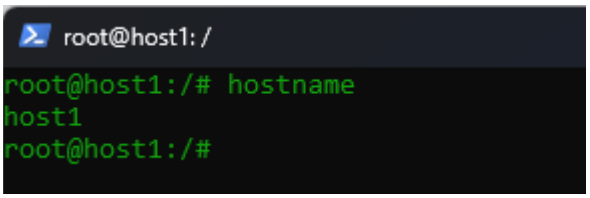
2º Instalamos ping:

apt update && apt install inetutils-ping -y

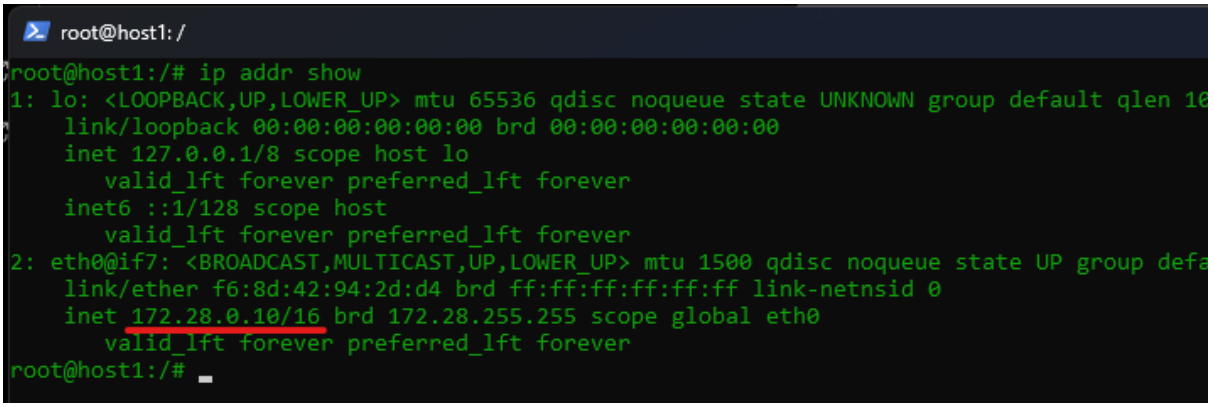


3º Comprobamos la configuración de red y el nombre de host:

hostname

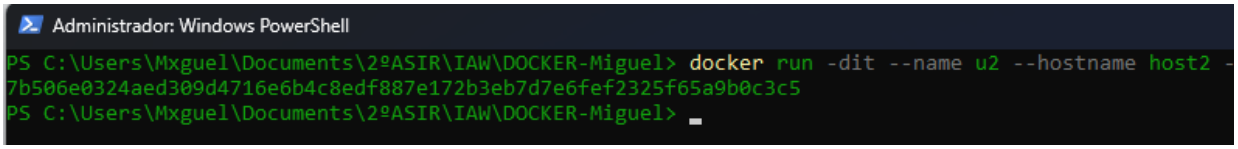


ip addr show



3. Creamos el segundo contenedor (u2)

docker run -dit --name u2 --hostname host2 --network red2 ubuntu:20.04



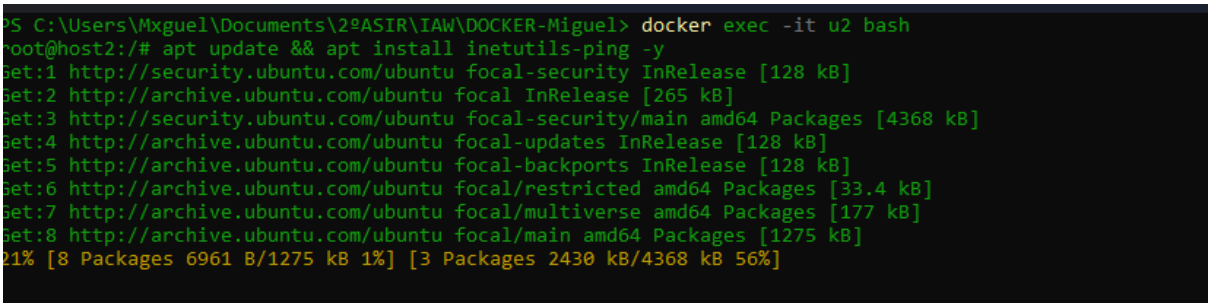
4. Instalamos ping en u2:

1º Lo ejecutamos:

docker exec -it u2 bash

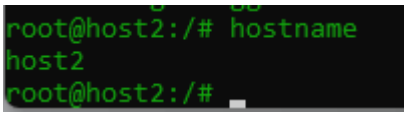
2º Instalamos ping:

apt update && apt install inetutils-ping -y

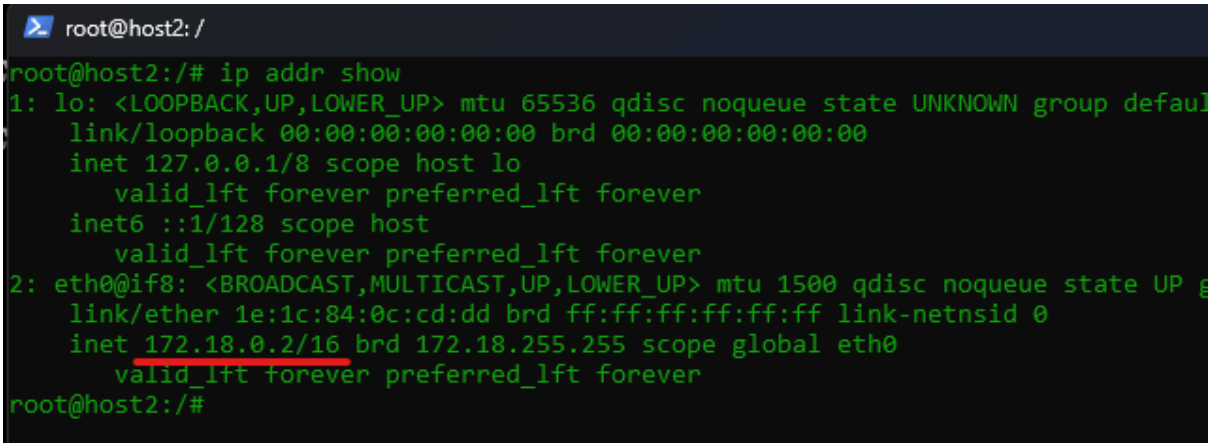


3º Comprobamos la configuración de red y el nombre de host:

hostname



ip addr show



5. Intentamos los pings

Por nombre de host:

ping host1

```
root@host2: /
root@host2:/# ping host1
ping: unknown host
```

ping host2

```
root@host1: /
root@host1:/# ping host2
ping: unknown host
```

Por IP:

ping 172.18.0.10 (IP u1)

```
root@host2:/# ping 172.28.0.10
PING 172.28.0.10 (172.28.0.10): 56 data bytes
^C--- 172.28.0.10 ping statistics ---
14 packets transmitted, 0 packets received, 100% packet loss
root@host2:/#
```

ping 172.18.0.2 (IP u2)

```
root@host1:/# ping 172.28.0.2
PING 172.28.0.2 (172.28.0.2): 56 data bytes
92 bytes from host1 (172.28.0.10): Destination Host Unreachable
92 bytes from host1 (172.28.0.10): Destination Host Unreachable
92 bytes from host1 (172.28.0.10): Destination Host Unreachable
92 bytes from host1 (172.28.0.10): Destination Host Unreachable
^C--- 172.28.0.2 ping statistics ---
6 packets transmitted, 0 packets received, 100% packet loss
root@host1:/#
```

6. Conectar u1 a la red2:

docker network connect red2 u1

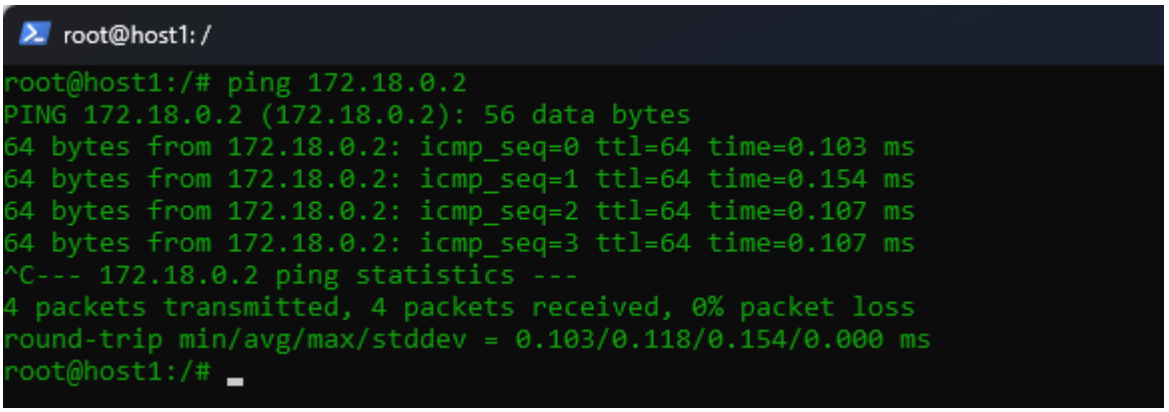
```
Administrador: Windows PowerShell
PS C:\Users\Mxguel\Documents\2ºASIR\IAW\DOCKER-Miguel> docker network connect red2 u1
```

7. Comprobamos que ahora sí se ven:

1º docker exec -it u1 bash

```
root@host1: /
root@host1:/# ping host2
PING host2 (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: icmp_seq=0 ttl=64 time=0.075 ms
64 bytes from 172.18.0.2: icmp_seq=1 ttl=64 time=0.113 ms
64 bytes from 172.18.0.2: icmp_seq=2 ttl=64 time=0.120 ms
64 bytes from 172.18.0.2: icmp_seq=3 ttl=64 time=0.114 ms
64 bytes from 172.18.0.2: icmp_seq=4 ttl=64 time=0.108 ms
64 bytes from 172.18.0.2: icmp_seq=5 ttl=64 time=0.115 ms
64 bytes from 172.18.0.2: icmp_seq=6 ttl=64 time=0.074 ms
```

2º ping 172.18.0.2 (IP u2)



Enlace a documentación GitHub:

https://github.com/Mxguell/DOCKER_MAY/tree/main/06-11-2026_TAREA%20MODULO%205